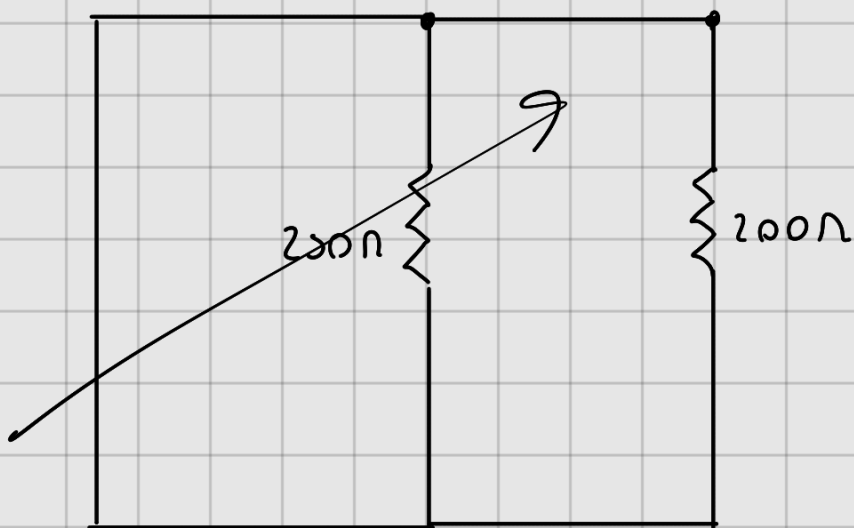
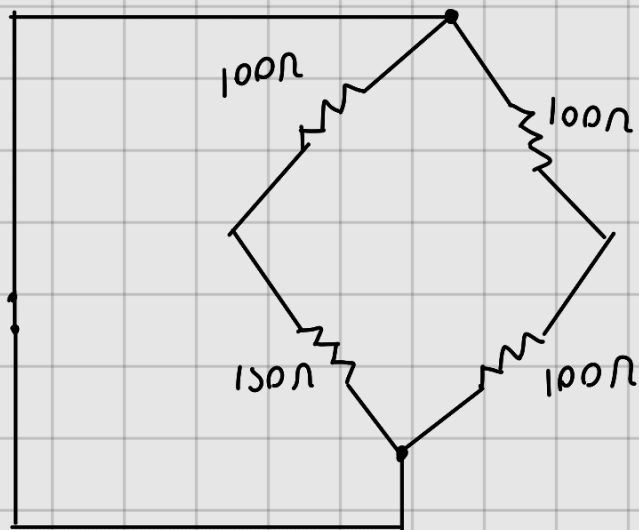
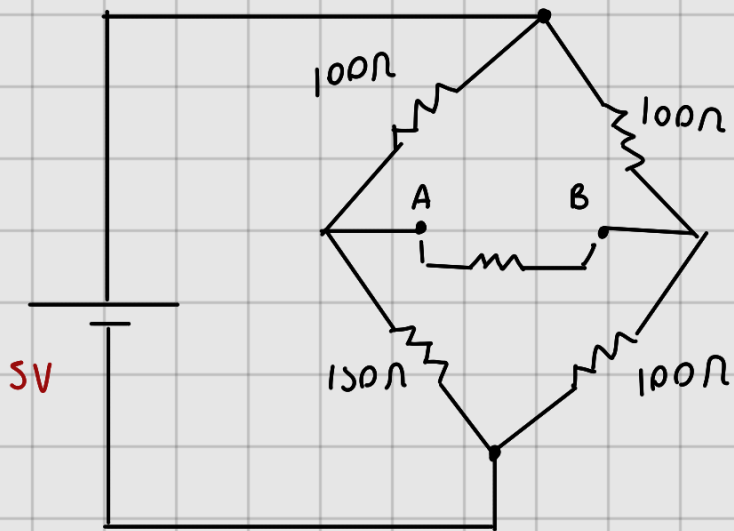
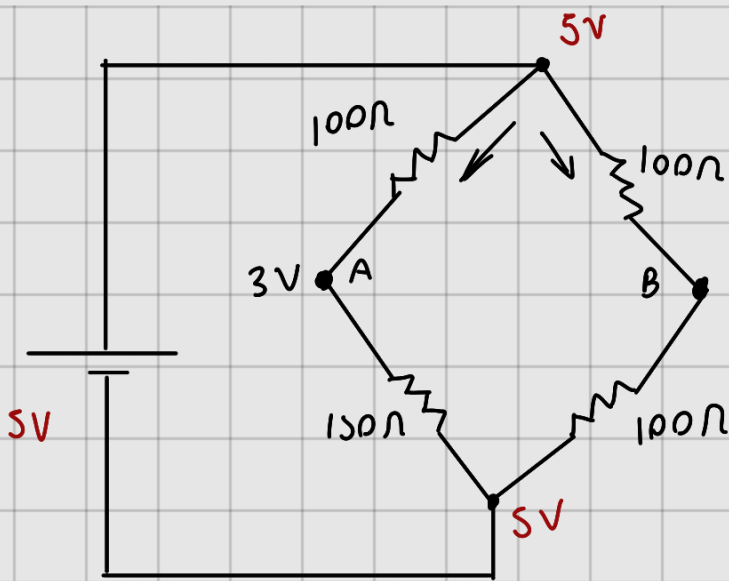




$$R_{TH} = 200\Omega$$



$$R_T = 111.111 \Omega$$

$$I_T = \frac{V_T}{R_T} = \frac{5V}{111.111}$$

$$I_T = 45 \text{ mA}$$

$$I_A = \frac{5V}{250\Omega} = 0.02 \text{ A} = 20 \text{ mA}$$

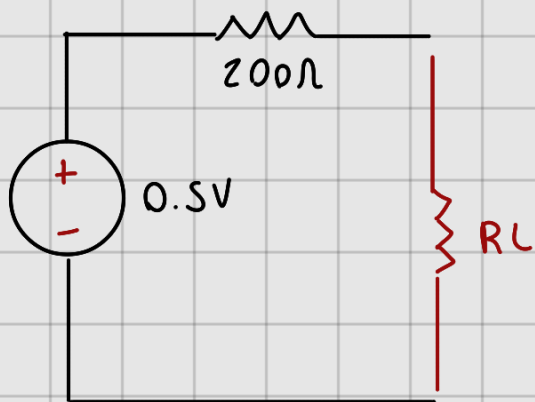
$$I_B = \frac{5V}{200\Omega} = 0.025 \text{ A} = 25 \text{ mA}$$

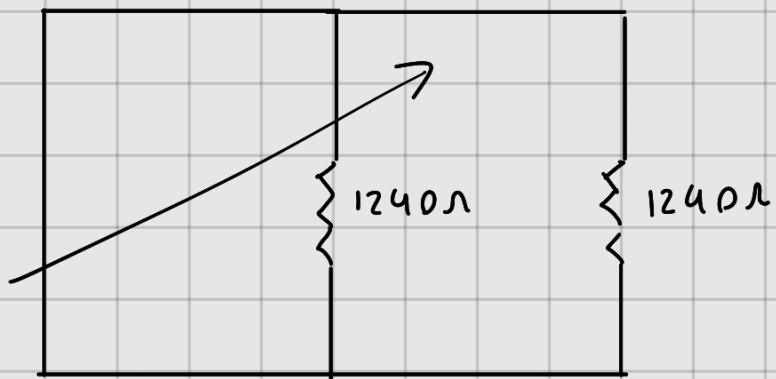
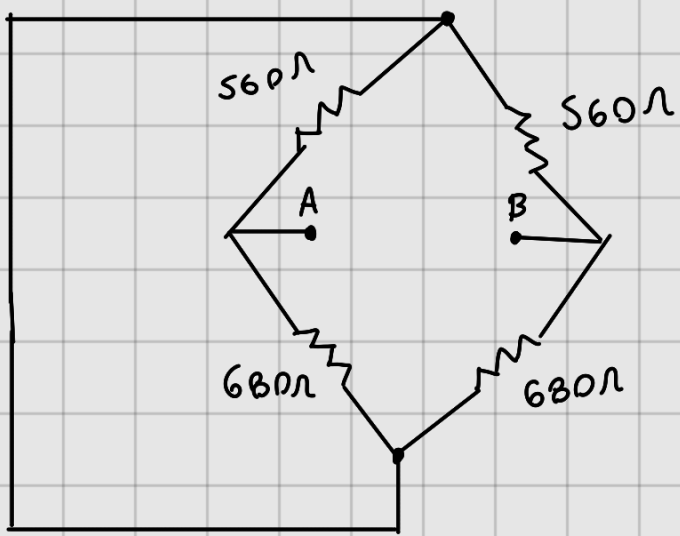
$$V_{100\Omega} = 100\Omega (20 \text{ mA}) = 2 \text{ V}$$

$$V_A = 150\Omega (20 \text{ mA}) = 3 \text{ V}$$

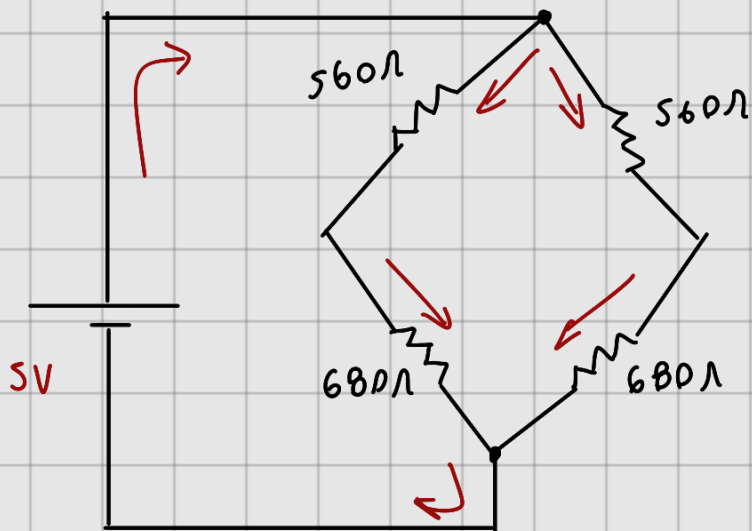
$$V_B = 100\Omega (25 \text{ mA}) = 2.5 \text{ V}$$

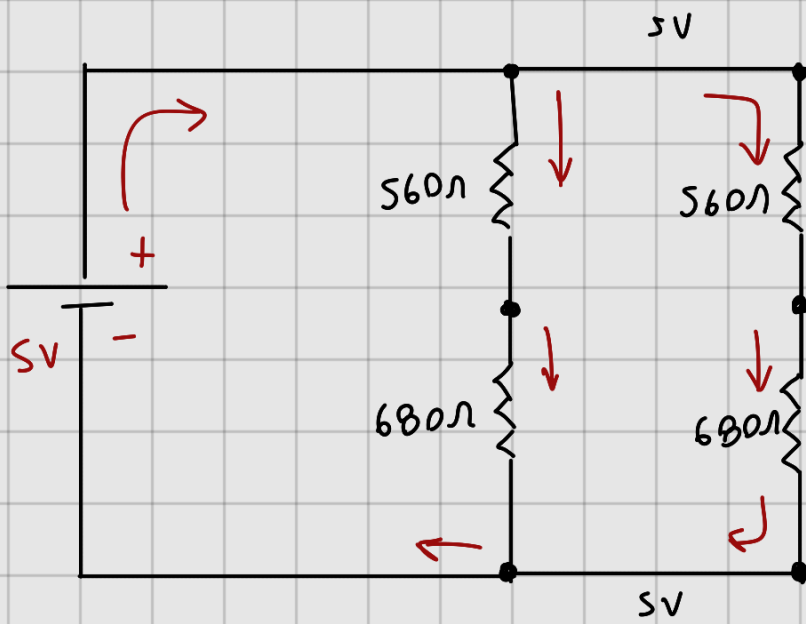
$$V_{TH} = 3 \text{ V} - 2.5 \text{ V} = 0.5 \text{ V}$$





$$R_N = 1240\Omega$$





$$R_T = 620\Omega$$

$$I_T = \frac{5V}{620\Omega} = 8.064mA$$

$$I_A = 4.032mA \rightarrow I_B$$

$$V_A = 2.74176 \quad V_B = 2.74176V$$

$$V_{TH} = 0V$$

